1 Two Changes to the Model

Two changes relative to the model section of the circulated draft matter for the calibration.

First, I drop the Stone–Geary structure of the draft, in which children entered through the committed consumption and housing bundles $(\bar{c}(n,s), \bar{h}(n,s))$, and model family size with an equivalence scale, the more common device in the quantitative lifecycle literature. An equivalence scale answers a simple question: how much more must a family spend than a childless couple to reach the same living standard. Consumption is then evaluated per adult equivalent, so a given bundle delivers less utility, and a marginal dollar more utility, in a larger household. Fernández-Villaverde and Krueger (2007) attribute about half of the lifecycle consumption hump to household size measured this way, and Scholz et al. (2006) and Borella et al. (2023) use scales of this form in lifecycle models close to this one. Within-period utility for a household with n dependent children at home is

$$u(c,h,n) = \frac{\left(c^{\alpha(n)}\, h^{1-\alpha(n)}/e(n)\right)^{1-\sigma}}{1-\sigma} + \psi\, n, \qquad \alpha(n) = \alpha_0 - \left(\delta_{\text{jump}}\mathbf{1}\{n \geq 1\} + \delta_a n\right), \tag{1}$$

Children raise the marginal value of the consumption–housing composite through $e(n)$[1] and tilt spending toward housing through $\alpha(n)$, both only while they live at home. Households without children at home have the homothetic preferences of the childless, and ψ is the direct utility flow per child at home. There are no minimum-consumption or minimum-housing requirements, which is what permits empirical income risk below.

Second, fertility is sequential rather than one-shot. Each period during the fertile ages, a childless household chooses between waiting and trying for a first birth, and a household with $n < 3$ children and a child still at home chooses between stopping and trying for child $n+1$. An attempt at age a succeeds with the fecundity probability

$$\pi_a = 1 - 0.02\, e^{0.134\,(a-18)} \quad \text{for } a < 45, \qquad \pi_a = 0 \text{ thereafter}, \tag{2}$$

which declines from 0.98 at age 18 to 0.50 at age 42.[2] Completed family size is therefore an outcome of starting age, biology, and sequential choices; no household ever chooses a number of children. Both decisions carry type-1 extreme-value taste shocks, with scale κ_E on the entry decision and κ_C on the continuation decision. The two scales are allowed to differ because the two decisions are different objects: the timing of the first attempt depends on circumstances the model does not describe, such as partnership timing and career events, so κ_E can be large, while the continuation decision of an established family should be governed by prices, income, and preferences, so κ_C is expected to be small.

2 Calibration

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of ten parameters estimated internally by matching twelve model moments to their empirical counterparts. I describe the two groups in turn, explain which moments discipline which parameters, and then report the target system.

[1] $e(n) = ((2 + 0.7n)/2)^{0.7}$, the scale of Citro and Michael (1995) used by Scholz et al. (2006), normalized to one for a childless couple.

[2] The two coefficients are set so that the four-year success probability tracks the age profile of natural fecundability in Leridon (2004). They imply four-year probabilities of about 0.90, 0.80, and 0.62 at ages 30, 35, and 40, against Leridon's estimates of 0.91, 0.84, and 0.64 for a conception leading to a live birth within four years. A least-squares fit to the full schedule is pending.

2.1 Parameters Set Outside the Estimation

Table 1 summarizes the parameters assigned outside the estimation. Demographics, housing finance, transaction costs, housing menus, and supply elasticity are unchanged from the circulated draft: a four-year period; entry, retirement, and maximum ages of 18, 66, and 85; survival from the 2023 Social Security period life table; an 18-year expected duration of dependent children; $\sigma = 2$; an annual real return of 2.0 percent, housing depreciation of 1.1 percent, property tax of 1.0 percent, selling cost of 6 percent, labor-income tax of 17.9 percent, and a financed share $\phi = 0.80$ (Greaney et al., 2025; Davis and Van Nieuwerburgh, 2015); owner sizes $\{2, 4, 6, 8, 10\}$ rooms and a 6-room rental cap from the matched ACS distributions; supply elasticity $\eta = 1.75$ (Saiz, 2010); and entrant wealth from the PSID distribution of childless renters aged 18–24.[3]

Four assigned objects are new relative to the draft.

The equivalence scale is fixed at the Citro–Michael form used by Scholz et al. (2006) and Borella et al. (2023); the square-root household-size scale is the robustness alternative.

The consumption share without children at home is fixed at $\alpha_0 = 0.733$. With Cobb–Douglas preferences and no committed bundles, the share of allocatable expenditure that a childless renter spends on housing is exactly $1 - \alpha_0$, so this parameter has a direct empirical counterpart: the CEX housing-expenditure share of childless cash renters in 2019–2023. A round that estimated α_0 freely returned 0.754, three percent from this external value.

The fecundity schedule is equation (2), anchored to the conception probabilities in Leridon's clinical tables with the formal least-squares fit pending. Family size takes the literal values $0, 1, 2, 3^+$, and the mean number of children in the top bin is fixed at 3.602, the capped mean among families with three or more children in the June 2024 CPS fertility supplement; the public file topcodes children ever born at five, and the model's completed-fertility moment uses the same capped convention, so model and data means share one definition.

The income process is set at its empirical value rather than truncated. The committed bundles of the previous specification forced income risk far below empirical estimates to keep the bundles affordable; without them the restriction disappears. I impose the persistent wage process of Flodén and Lindé (2001), with annual persistence 0.9136 and innovation standard deviation 0.206 net of measurement error, passed through the log-linear tax function of Heathcote et al. (2017) with progressivity $\tau = 0.181$, which preserves the autoregressive structure and scales the after-tax innovation to 0.169. The process is sampled at the four-year model frequency and approximated with a five-state Rouwenhorst chain. Unlike in the draft, this object is no longer provisional, and the draft's footnote conditioning all quantitative results on a truncated income variance no longer applies.

[3] Bequest utility does not vary directly with the number of children.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period; entry, retirement, maximum ages	4 years; 18, 66, 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Equivalence scale $e(n)$	$((2+0.7n)/2)^{0.7}$	Scholz et al. (2006), from Citro and Michael (1995)
Consumption share α_0	0.733	CEX housing-expenditure share, childless cash renters, 2019–2023
Fecundity by age	Eq. (2); zero from 45	Leridon (2004); formal fit pending
Family-size states	$0, 1, 2, 3^+$; top-bin mean 3.602	CPS June 2024 capped mean among 3^+ families
Annual income persistence ρ_z	0.9136	Flodén and Lindé (2001)
Annual income innovation s.d. σ_z	0.169	Flodén and Lindé (2001) innovation 0.206, scaled by $1-\tau$ with $\tau = 0.181$ from Heathcote et al. (2017)
Annual real return; depreciation; property tax	2.0%; 1.1%; 1.0%	Greaney et al. (2025)
Financed share ϕ; sale cost ψ^s	0.80; 6%	Davis and Van Nieuwerburgh (2015)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average
Owner house sizes; largest rental unit	$\{2, 4, 6, 8, 10\}$; 6 rooms	Matched ACS distributions
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income draws	PSID childless renters aged 18–24

2.2 Internally Estimated Parameters

I estimate the remaining ten parameters internally against twelve moments. Table 2 reports the current estimates. The parameters are chosen jointly. I group them into four economic blocks and, for each, describe which moments are most informative.

Table 2: Internally estimated parameters (current diagnostic estimates; the estimation on the revised target system is running)

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9705
δ_{jump}	Expenditure-share shift at entry into parenthood	0.0373
δ_a	Expenditure-share shift per additional child	0.0242
ψ	Utility flow per child at home	2.8137
κ_E	Taste-shock scale, first-birth entry decision	49.38
κ_C	Taste-shock scale, continuation decisions	0.1806
χ_O	Utility premium from owner-occupied housing	1.0588
$\bar{H}$	Housing-supply scale	8.5226
θ_0	Strength of the bequest motive	0.0264
θ_1	Bequest curvature shift	0.0442

Fertility. The utility value of children ψ raises the return to every birth, so it governs how many children households have; completed fertility is most informative about it. The entry noise κ_E spreads the age at which women start trying, and biology turns late starts into late or missing births, so it governs the timing of first births; the mean age at first birth and the share of first births after 30 are most informative about it. The continuation noise κ_C governs how readily a household that has started keeps going: when it is small, starting a family is close to a commitment to several children, so fewer women start at all; childlessness is most informative about it. The three are estimated together, because the same fertility outcomes respond to all of them, and the progression rates between birth orders are left untargeted. The timing moments are new to this specification. Without them the entry noise is pinned only through the level of fertility, and it drifts high enough that first births barely respond to prices, which understates the housing channel the policy experiments are meant to measure.

Family housing. The two shifters δ_{jump} and δ_a set how far a household tilts spending toward housing when children arrive. The rooms a household adds at the first child is most informative about the total tilt at entry into parenthood, $\delta_{\text{jump}} + \delta_a$, since a first-time parent takes both the jump and one per-child step; the rooms difference between families with three or more children and families with one or two is most informative about δ_a alone, since the jump is common to both and drops out. The family ownership gap has no shifter of its own. It responds to the tilts and to who remains childless, and I keep it in the estimation because leaving it out lets the model deliver a gap of 0.04 to 0.07 against 0.168 in the data.

Tenure and supply. The owner premium χ_O raises the utility of owning over renting; the prime-age ownership rate is most informative about it. The supply scale $\bar{H}$ sets the level of the housing stock; aggregate occupied rooms is most informative about it. Tenure choice is deterministic.[4]

Saving and bequests. The discount factor β_{annual} governs how much wealth households carry through life; aggregate net worth relative to annual gross labor earnings, 6.873 in the PSID, is most informative about it. The bequest strength θ_0 governs how much households hold back to leave at death; the annual flow of bequests relative to aggregate wealth, 0.0088, is most informative about it.[5] The bequest curvature θ_1 governs how the motive strengthens with wealth; the ratio of the 90th to the 50th percentile of total wealth late in life, 3.448, is most informative about it. Wealth is measured on the beginning-of-period balance sheet, so liquid wealth and housing refer to the same date, and the bequest is measured at death after the period's saving. The flow moment matters for tenure late in life: without it the bequest motive falls away, nothing rewards holding wealth to death, and older owners never sell, so ownership after 65 climbs to 0.945.

2.3 Target Moments and Fit

The four family-size and timing moments are measured for the 1979–1984 birth cohorts: completed fertility and childlessness for women aged 40–44 in the June 2024 CPS fertility supplement, and the two first-birth timing moments from the NCHS natality microdata for mothers of the same cohorts (10.2 million first births; the observed truncation of the timing distribution is about one percent). These are the youngest cohorts with essentially complete fertility. The environment (prices, incomes, entrant

[4] A diagnostic round that added a taste shock to the own–rent margin drove its scale to zero at no cost to fit: empirical income risk moves households between owning and renting without taste noise. The four-year tenure switch rate is kept as an untargeted check on the income process.

[5] The flow is a borrowed number, from Gale and Scholz (1994) through De Nardi and Yang (2014).

wealth, housing menus) is measured on 2019–2023 cross sections; holding preferences fixed across these adjacent cohorts is a maintained assumption, and a robustness package re-targeting current period fertility rates is planned.

Each moment enters the loss with weight equal to the inverse of its squared standard error, so the loss reads as squared standard errors of distance. Bootstrap standard errors are used where the target is project-measured (completed fertility 0.026, childlessness 0.008, the wealth ratio 0.399, the dispersion ratio 0.133); the timing rows carry declared working values (0.25 years and 0.008) pending the cohort-table bootstrap; the remaining rows carry five percent of the target value.

Table 3: Target moments (model column pending the revised-system estimation)

Moment	Source	Data	Model
Family size and timing			
Completed fertility	CPS June 2024, women 40–44	1.918	—
Childlessness rate	CPS June 2024, women 40–44	0.188	—
Mean age at first birth	NCHS natality, 1979–84 cohorts	25.31	—
Share of first births at ages 30+	NCHS natality, 1979–84 cohorts	0.270	—
Family housing and tenure			
Housing increment with the first child, rooms	PSID event study	0.664	—
Rooms gap, 3+ versus 1–2 children, ages 30–55	ACS	0.368	—
Family ownership gap, ages 30–55	ACS	0.168	—
Ownership rate, ages 30–55	ACS	0.575	—
Mean occupied rooms, ages 18–85	ACS	5.780	—
Wealth and bequests			
Aggregate wealth / annual gross labor earnings	PSID 2005–2019	6.873	—
Annual bequest flow / aggregate wealth	Gale and Scholz (1994)	0.0088	—
p_{90}/p_{50} total wealth, ages 76–84	PSID	3.448	—

A diagnostic estimate on the previous fifteen-moment system shows what the revision builds on. The sequential model with two noise scales reaches completed fertility of 1.875 and childlessness of 0.155, against 1.918 and 0.188 in the data, a pair no earlier specification reached without overshooting fertility; about half of the model's childlessness comes from women who try and run out the fertile window, the postponement mechanism at the center of the paper. That estimate also showed where the moment system was too thin: the entry noise was loose without a timing moment, the bequest parameters were loose without a flow moment, and the family ownership gap came in too small. The revised estimation is running as I write; the model column above, the estimates in Table 2, and the figure below will follow when it finishes.

Figure 1 compares model homeownership, the share of households with dependent children at home, and mean rooms with their ACS age profiles at the current estimate.

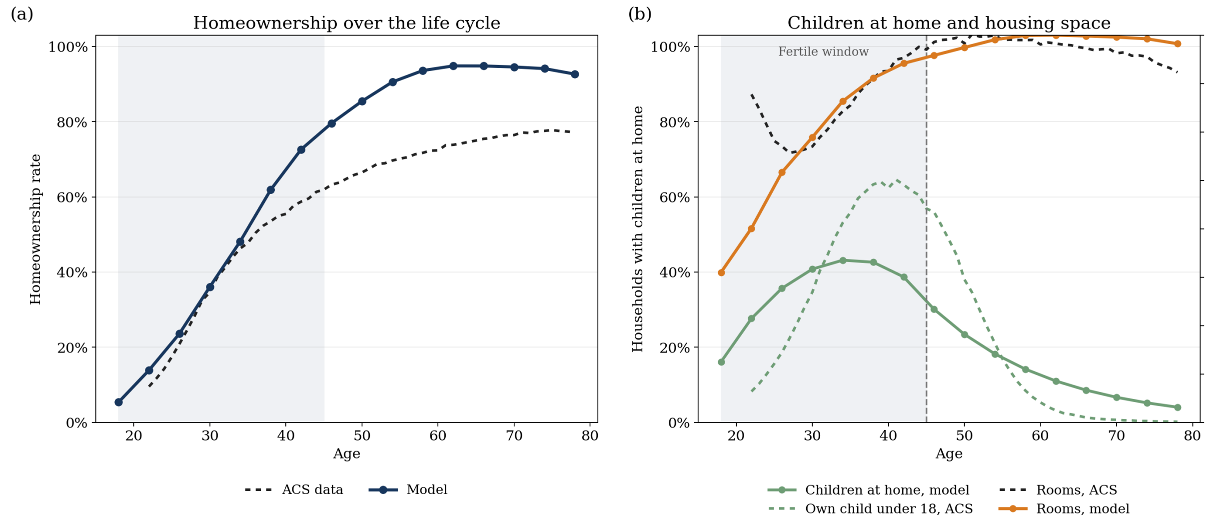

Figure 1: Lifecycle ownership, children at home, and housing consumption at the current estimate. Panel (a) compares model homeownership with the ACS household-head profile. Panel (b) compares the model share of households with dependent children at home with the ACS share of household heads with an own child under age 18, and compares mean rooms in the model and the ACS. The shaded region marks ages 18–45, the fertile window.

References

Borella, M., M. De Nardi, and F. Yang. 2023. "Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labour Supply?" *Review of Economic Studies* 90(1): 102–131.

Davis, M. A., and S. Van Nieuwerburgh. 2015. "Housing, Finance, and the Macroeconomy." *Handbook of Regional and Urban Economics* 5: 753–811.

De Nardi, M., and F. Yang. 2014. "Bequests and Heterogeneity in Retirement Wealth." *European Economic Review* 72: 182–196.

Fernández-Villaverde, J., and D. Krueger. 2007. "Consumption over the Life Cycle: Facts from Consumer Expenditure Survey Data." *Review of Economics and Statistics* 89(3): 552–565.

Flodén, M., and J. Lindé. 2001. "Idiosyncratic Risk in the United States and Sweden: Is There a Role for Government Insurance?" *Review of Economic Dynamics* 4(2): 406–437.

Gale, W. G., and J. K. Scholz. 1994. "Intergenerational Transfers and the Accumulation of Wealth." *Journal of Economic Perspectives* 8(4): 145–160.

Greaney, B., A. Parkhomenko, and S. Van Nieuwerburgh. 2025. "Dynamic Urban Economics." Working paper.

Heathcote, J., K. Storesletten, and G. L. Violante. 2017. "Optimal Tax Progressivity: An Analytical Framework." *Quarterly Journal of Economics* 132(4): 1693–1754.

Leridon, H. 2004. "Can Assisted Reproduction Technology Compensate for the Natural Decline in Fertility with Age? A Model Assessment." *Human Reproduction* 19(7): 1548–1553.

Saiz, A. 2010. "The Geographic Determinants of Housing Supply." *Quarterly Journal of Economics* 125(3): 1253–1296.

Scholz, J. K., A. Seshadri, and S. Khitatrakun. 2006. "Are Americans Saving 'Optimally' for Retirement?" *Journal of Political Economy* 114(4): 607–643.